Chapter 1

ELEMENTARY LOGIC

EXERCISE 1.1

Multiple Choice Questions (MCQs)

Q.1 **Tick ✓ the correct option in the following questions:**

(i) **Which of the following is a proposition?**

(a) What is your age? **(b) Always work hard.**
(c) Do not irritate him **(d) Quetta is a city of Pakistan**

Sol: (a), (b) and (c) are not propositions, because these are neither true nor false.
(d) is a proposition and its truth value is true (T).
Correct answer is option (d).

(ii) **Which of the following is not a proposition?**

(a) $7 > 9$ **(b) $2 + 3 = 5$** **(c) $x < 5$** **(d) $2 + 3 \neq 5$**

Sol: (c) is not a proposition, because it is neither true nor false.
Correct answer is option (c).

(iii) **Which of the following is a proposition?**

(a) Do not smoke **(b) $x + y = z$**
(c) Lahore is capital of Punjab **(d) Walk along the road**

Sol: (c) is a proposition and its truth value is true (T).
Correct answer is option (c).

(iv) **The negation of "Today is Friday" is**

(a) Today is Saturday **(b) Today is not Friday**
(c) Today is Thursday **(d) None of these**

Sol: The negation of "Today is Friday" is "Today is not Friday"
Correct answer is option (b).

(v) **If $p : x + y = 3$, then $\neg p$ is**

(a) $x + y \neq 3$ **(b) $x + y > 3$** **(c) $x + y < 3$** **(d) $x + y = 0$**

Sol: The negation of "$x + y = 3$" is "$x + y \neq 3$"
Correct answer is option (a).

(vi) **An arrangement of rows and columns that specifies the truth value of a compound proposition for all possible truth values of its constituent propositions is called**

(a) Truth Table **(b) Venn diagram**
(c) False Table **(d) None of these**

Sol: Correct answer is option (a).

(vii) **Contrapositive of given statement "If it is raining, I will take an umbrella" is**

(a) I will not take an umbrella if it is not raining
(b) I will take an umbrella if it is raining
(c) It is not raining or I will take an umbrella
(d) None of these

Sol: Correct answer is option (a).

(viii) **$p \wedge q$ is true if**

(a) p is false **(b) q is false**
(c) both p and q are false **(d) both p and q are true**

Sol: $p \wedge q$ is true if both p and q are true
Correct answer is option (d).

(ix) **$p \vee q$ is false if**

(a) p is false **(b) q is false**
(c) both p and q are false **(d) both p and q are true**

Sol: $p \vee q$ is false if both p and q are false
Correct answer is option (c).

(x) **$p \oplus q$ is true if**

(a) p is false **(b) q is false and p is true**
(c) both p and q are false **(d) both p and q are true**

Sol: $p \oplus q$ is the proposition that is true when exactly one of p and q is true and is false otherwise.
Correct answer is option (b).

(xi) **If $p : 2^2 > 1^2$; q : every odd number is divisible by 2, then**

(a) $p \vee q$ is false **(b) $p \vee q$ is true**
(c) $p \wedge q$ is true **(d) $p \oplus q$ is false**

Sol: Here p is true and q is false, so $p \vee q$ is true.
Correct answer is option (b).

(xii) **If $p : 2 + 3 = 5$; q : every even number is divisible by 2, then**

(a) $p \vee q$ is false **(b) $p \wedge q$ is false**
(c) $p \wedge q$ is true **(d) $p \oplus q$ is true**

Sol: Here both p and q are true, so $p \wedge q$ is true.
Correct answer is option (c).

(xiii) **If p : Islamabad is a capital of Pakistan; q : No even integer is divisible by 2, then which of the following is true?**

(a) $p \wedge q$ **(b) $p \oplus q$** **(c) $p \oplus \neg q$** **(d) $\neg p \vee q$**

Sol: Here p is true and q is false.
Correct answer is option (b).

(xiv) If p : no lion eats meat; q : 1 + 1 = 2, then

(a) $p \to q$ is false (b) $p \to q$ is true

(c) $p \oplus \neg q$ is true (d) $\neg p \wedge q$ false

Sol: Here p is false and q is true.
Correct answer is option (b).

(xv) The converse of $p \to q$ is

(a) $q \to p$ (b) $\neg p \to \neg q$

(c) $\neg q \to \neg p$ (d) $p \leftrightarrow q$

Sol: Correct answer is option (a).

(xvi) The inverse of $p \to q$ is

(a) $q \to p$ (b) $\neg p \to \neg q$

(c) $\neg q \to \neg p$ (d) $p \leftrightarrow q$

Sol: Correct answer is option (b).

(xvii) The contrapositive of $p \to q$ is

(a) $q \to p$ (b) $\neg p \to \neg q$

(c) $\neg q \to \neg p$ (d) $p \leftrightarrow q$

Sol: Correct answer is option (c).

Short Questions

Solve / write answers of the following short questions:

Q.2 Which of the following sentences are propositions? What are the truth values of those that are propositions?

(i) **Do not tell lie.**

Sol: This is not a proposition, because it is neither true nor false.

(ii) **Always speak truth.**

Sol: This is not a proposition, because it is neither true nor false.

(iii) **Chicago is situated in USA.**

Sol: This is a proposition and its truth value is true (T).

(iv) $4^2 + 3^2 = 5^2$

Sol: This is a proposition and its truth value is true (T).

(v) $18 > 13$

Sol: This is a proposition and its truth value is true (T).

(vi) $17 + 3 < 9$

Sol: This is a proposition and its truth value is false (F).

(vii) $x + 1 > 5$

Sol: This is not a proposition, because it is neither true nor false.

(viii) **Product of two negative real numbers is a positive real number.**

Sol: This is a proposition and its truth value is true (T).

(ix) **Do not waste your time.**

Sol: This is not a proposition, because it is neither true nor false.

(x) **Cats are birds.**

Sol: This is a proposition and its truth value is false (F).

Q.3 What is the negation of each of these propositions?

(i) There is pollution in Lahore.

Sol: There is no pollution in Lahore.

(ii) Umair and Kashif are friends.

Sol: Umair and Kashif are not friends.

(iii) Babar sent more than 100 text messages last week.

Sol: Babar did not send more than 100 text messages last week.

(iv) Madiha has no MP3 player.

Sol: Madiha has MP3 player.

(v) 64 is a perfect square.

Sol: 64 is not a perfect square.

(vi) Milkman was caught while mixing water in milk.

Sol: Milkman was not caught while mixing water in milk.

(vii) Elephants eat sugarcanes.

Sol: Elephants do not eat sugarcanes.

(viii) Education is necessary for women.

Sol: Edı cation is not necessary for women.

(ix) $2^3 + 1^3 = 3^2$.

Sol: $2^3 + 1^3 \neq 3^2$.

(x) Shazia has more than 150 GB free disk space on her laptop.

Sol: Shazia has less than 150 GB free disk space on her laptop

Q.4 If p is any proposition, then construct the following truth tables.

(i) $p \wedge \neg p$ (ii) $p \vee \neg p$

(iii) $p \oplus \neg p$ (iv) $p \to \neg p$

(v) $\neg p \to p$ (vi) $p \leftrightarrow \neg p$

(vii) $p \oplus p$ (viii) $p \wedge p$

(ix) $(p \wedge \neg p) \oplus p$ (x) $(p \vee \neg p) \oplus p$

Sol:(i) $p \wedge \neg p$

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

(ii) $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

(iii) $p \oplus \neg p$

p	$\neg p$	$p \oplus \neg p$
T	F	T
F	T	T

(iv) $p \to \neg p$

p	$\neg p$	$p \to \neg p$
T	F	F
F	T	T

(v) $\neg p \to p$

p	$\neg p$	$\neg p \to p$
T	F	T
F	T	F

(vi) $p \leftrightarrow \neg p$

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

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(vii) $p \oplus p$

p	$p \oplus p$
T	F
F	F

(viii) $p \wedge p$

p	$p \wedge p$
T	T
F	F

(ix) $(p \wedge \neg p) \oplus p$

p	$\neg p$	$p \wedge \neg p$	$(p \wedge \neg p) \oplus p$
T	F	F	T
F	T	F	F

(x) $(p \vee \neg p) \oplus p$

p	$\neg p$	$p \vee \neg p$	$(p \vee \neg p) \oplus p$
T	F	T	F
F	T	T	T

Q.5 Determine whether each of the following conditionals is true or false.

(i) If 1 + 1 = 3, then cats can fly.

Sol: Let

$p : 1 + 1 = 3;$ q : cats can fly

Then the conditional becomes

$p \rightarrow q$: If 1 + 1 = 3, then cats can fly.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
F	F	T

This shows that the truth value of conditional

"If 1 + 1 = 3, then cats can fly"

is true.

(ii) If 1 + 2 = 3, then cats can fly.

Sol: Let

$p : 1 + 2 = 3;$ q : cats can fly

Then the conditional becomes

$p \rightarrow q$: If 1 + 2 = 3, then cats can fly.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
T	F	F

This shows that the truth value of conditional

"If 1 + 2 = 3, then cats can fly"

is false.

(iii) If 3 < 1, then 2 < 1.

Sol: Let

$p : 3 < 1$; $q : 2 < 1$

Then the conditional becomes

$p \rightarrow q$: If $3 < 1$, then $2 < 1$.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
F	F	T

This shows that the truth value of conditional

"If $3 < 1$, then $2 < 1$"

is true.

(iv) **If $3 < 1$, then $1 < 2$.**

Sol: Let

$p : 3 < 1$; $q : 1 < 2$

Then the conditional becomes

$p \rightarrow q$: If $3 < 1$, then $1 < 2$.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
F	T	T

This shows that the truth value of conditional

"If $3 < 1$, then $1 < 2$"

is true.

(v) **If $3 > 1$, then $2 < 1$.**

Sol: Let

$p : 3 > 1$; $q : 2 < 1$

Then the conditional becomes

$p \rightarrow q$: If $3 > 1$, then $2 < 1$.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
T	F	F

This shows that the truth value of conditional

"If $3 > 1$, then $2 < 1$"

is false.

(vi) **If $3 + 3 = 6$, then $1 < 1$.**

Sol: Let

$p : 3 + 3 = 6$; $q : 1 < 1$

Then the conditional becomes

$p \rightarrow q$: If $3 + 3 = 6$, then $1 < 1$.

Next consider the truth table of conditional.

p	q	$p \rightarrow q$
T	F	F

This shows that the truth value of conditional

"If $3 + 3 = 6$, then $1 < 1$"

is false.

(vii) **If monkeys can fly, then sparrows cannot fly.**

Sol: Let

p : Monkeys can fly; q : Sparrows cannot fly

Then the conditional becomes

$p \to q$: If monkeys can fly, then sparrows cannot fly.

Next consider the truth table of conditional.

p	q	$p \to q$
F	F	T

This shows that the truth value of conditional

"If monkeys can fly, then sparrows cannot fly."

is true.

(viii) **If sparrows can fly, then monkeys cannot fly.**

Sol: Let

p : Sparrows can fly; q : Monkeys cannot fly

Then the conditional becomes

$p \to q$: If sparrows can fly, then monkeys cannot fly.

Next consider the truth table of conditional.

p	q	$p \to q$
T	T	T

This shows that the truth value of conditional

"If sparrows can fly, then monkeys cannot fly."

is true.

(ix) **If a car can travel without fuel, then a plane can fly without fuel.**

Sol: Let

p : A car can travel without fuel;

q : A plane can fly without fuel

Then the conditional becomes

$p \to q$: If a car can travel without fuel, then a plane can fly without fuel.

Next consider the truth table of conditional.

p	q	$p \to q$
F	F	T

This shows that the truth value of conditional

"If a car can travel without fuel, then a plane can fly without fuel."

is true.

(x) **If fish can live without water, then lion can breathe in water.**

Sol: Let

p : Fish can live without water;
q : Lion can breathe in water
Then the conditional becomes
$p \to q$: If fish can live without water, then lion can breathe in water.
Next consider the truth table of conditional.

p	q	$p \to q$
F	F	T

This shows that the truth value of conditional
"If fish can live without water, then lion can breathe in water."
is true.

Q.6 Determine whether the following biconditionals are true or false.

(i) $3 + 2 = 5$ if and only if $6 + 1 = 7$.

Sol: Let
$p : 3 + 2 = 5$; $q : 6 + 1 = 7$
Then the given biconditional becomes
$p \leftrightarrow q : 3 + 2 = 5$ if and only if $6 + 1 = 7$.
Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
T	T	T

This shows that the truth value of biconditional
"$3 + 2 = 5$ if and only if $6 + 1 = 7$"
is true.

(ii) $3 + 2 \neq 5$ if and only if $6 + 1 = 7$.

Sol: Let
$p : 3 + 2 \neq 5$; $q : 6 + 1 = 7$
Then the given biconditional becomes
$p \leftrightarrow q : 3 + 2 \neq 5$ if and only if $6 + 1 = 7$.
Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	T	F

This shows that the truth value of biconditional
"$3 + 2 \neq 5$ if and only if $6 + 1 = 7$"
is false.

(iii) $3 + 2 \neq 5$ if and only if $6 + 1 \neq 7$.

Sol: Let
$p : 3 + 2 \neq 5$; $q : 6 + 1 \neq 7$
Then the given biconditional becomes
$p \leftrightarrow q : 3 + 2 \neq 5$ if and only if $6 + 1 \neq 7$.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	F	T

This shows that the truth value of biconditional

"$3 + 2 \neq 5$ if and only if $6 + 1 \neq 7$"

is true.

(iv) 3 + 1 = 7 if and only if dogs can fly.

Sol: Let

$p : 3 + 1 = 7$; q : Dogs can fly

Then the given biconditional becomes

$p \leftrightarrow q : 3 + 1 = 7$ if and only if dogs can fly.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	F	T

This shows that the truth value of biconditional

"$3 + 1 = 7$ if and only if dogs can fly"

is true.

(v) Lahore is in Punjab if and only if Chicago is in USA.

Sol: Let

p : Lahore is in Punjab; q : Chicago is in USA

Then the given biconditional becomes

$p \leftrightarrow q$: Lahore is in Punjab if and only if Chicago is in USA.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
T	T	T

This shows that the truth value of biconditional

"Lahore is in Punjab if and only if Chicago is in USA."

is true.

(vi) Pakistan is in Asia if and only if New Delhi is Pakistan.

Sol: Let

p : Pakistan is in Asia; q : New Delhi is Pakistan

Then the given biconditional becomes

$p \leftrightarrow q$: Pakistan is in Asia if and only if New Delhi is Pakistan.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
T	F	F

This shows that the truth value of biconditional

"Pakistan is in Asia if and only if New Delhi is Pakistan."

is false.

(vii) **0 > 1 if and only if 3 > 1.**

Sol: Let

$p : 0 > 1$; $q : 3 > 1$

Then the given biconditional becomes

$p \leftrightarrow q$: 0 > 1 if and only if 3 > 1.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	T	F

This shows that the truth value of biconditional

"0 > 1 if and only if 3 > 1"

is false.

(viii) **0 > 1 if and only if 3 < 1.**

Sol: Let

$p : 0 > 1$; $q : 3 < 1$

Then the given biconditional becomes

$p \leftrightarrow q$: 0 > 1 if and only if 3 < 1.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	F	T

This shows that the truth value of biconditional

"0 > 1 if and only if 3 < 1"

is true.

(ix) **7 + 8 = 15 if and only if 2 > 5.**

Sol: Let

$p : 7 + 8 = 15$; $q : 2 > 5$

Then the given biconditional becomes

$p \leftrightarrow q$: 7 + 8 = 15 if and only if 2 > 5.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
T	F	F

This shows that the truth value of biconditional

"7 + 8 = 15 if and only if 2 > 5"

is false.

(x) **Monkeys can fly if and only sparrows can fly.**

Sol: Let

p : Monkeys can fly; q : Sparrows can fly

Then the given biconditional becomes

$p \leftrightarrow q$: Monkeys can fly if and only sparrows can fly.

Next consider the truth table of biconditional.

p	q	$p \leftrightarrow q$
F	T	F

This shows that the truth value of biconditional
"Monkeys can fly if and only sparrows can fly."
is false.

Q.7 What is the value of x after each of the following statements is encountered in a computer program, if $x = 1$ before the statement is reached?

(i) if $x + 3 = 3$ then $x := x + 1$

Sol: For $x = 1$, we have $x + 3 = 1 + 3 = 4 \neq 3$
This shows that, for $x = 1$, $x + 3 = 3$ is false.
The statement does not execute, and the value of x remains at is, i.e. $x = 1$.

(ii) if $x + 3 = 4$ then $x := x + 1$

Sol: For $x = 1$, we have $x + 3 = 1 + 3 = 4$
This shows that, for $x = 1$, $x + 3 = 4$ is true.
The statement executes, and the value of x changes from x to $x + 1$, i.e. $x = 2$.

(iii) if $(2x + 1 = 3) \vee (x + 2 = 4)$ then $x := x + 2$

Sol: For $x = 1$, $(2x + 1 = 3) \vee (x + 2 = 4)$ becomes
$(2 + 1 = 3) \vee (1 + 2 = 4)$
This shows that, for $x = 1$, $(2x + 1 = 3) \vee (x + 2 = 4)$ is true.
The statement executes, and the value of x changes from x to $x + 2$, i.e. $x = 3$.

(iv) if $(2x + 1 = 3) \wedge (x + 2 = 4)$ then $x := x + 2$

Sol: For $x = 1$, $(2x + 1 = 3) \wedge (x + 2 = 4)$ becomes
$(2 + 1 = 3) \wedge (1 + 2 = 4)$
This shows that, for $x = 1$, $(2x + 1 = 3) \wedge (x + 2 = 4)$ is false.
The statement does not execute, and the value of x does not change, so $x = 1$.

(v) if $(2x + 3 = 5) \wedge (3x + 4 = 7)$ then $x := x + 7$

Sol: For $x = 1$, $(2x + 3 = 5) \wedge (3x + 4 = 7)$ becomes
$(2 + 3 = 5) \wedge (3 + 4 = 7)$
This shows that, for $x = 1$, $(2x + 3 = 5) \wedge (3x + 4 = 7)$ is true.
The statement executes, and the value of x changes from x to $x + 7$, i.e. $x = 8$.

(vi) if $(2x + 3 = 5) \vee (3x + 4 = 7)$ then $x := x + 7$

Sol: For $x = 1$, $(2x + 3 = 5) \vee (3x + 4 = 7)$ becomes
$(2 + 3 = 5) \vee (3 + 4 = 7)$
This shows that, for $x = 1$, $(2x + 3 = 5) \vee (3x + 4 = 7)$ is true.
The statement executes, and the value of x changes from x to $x + 7$, i.e. $x = 8$.

(vii) if $(x + 2 = 3) \oplus (x + 2 \neq 3)$ then $x := x + 1$

Sol: For $x = 1$, $(x + 2 = 3) \oplus (x + 2 \neq 3)$ becomes
$(1 + 2 = 3) \oplus (1 + 2 \neq 3)$
This shows that, for $x = 1$, $(x + 2 = 3) \oplus (x + 2 \neq 3)$ is true.
The statement executes, and the value of x changes from x to $x + 1$, i.e. $x = 2$.

(viii) If $(x + 2 = 3) \oplus (x + 2 = 2)$ then $x := x + 1$

Sol: For $x = 1$, $(x + 2 = 3) \oplus (x + 2 = 2)$ becomes
$(1 + 2 = 3) \oplus (1 + 2 = 2)$
This shows that, for $x = 1$, $(x + 2 = 3) \oplus (x + 2 = 2)$ is true.
The statement executes, and the value of x changes from x to $x + 1$, i.e. $x = 2$.

(ix) If $x < 3$ then $x := x + 3$

Sol: For $x = 1$, $x < 3$ becomes $1 < 3$
This shows that, for $x = 1$, $x < 3$ is true.
The statement executes, and the value of x changes from x to $x + 3$, i.e. $x = 4$.

(x) If $x > 3$ then $x := x + 3$

Sol: For $x = 1$, $x > 3$ becomes $1 > 3$
This shows that, for $x = 1$, $x > 3$ is false.
The statement does not execute, and the value of x does not change, so $x = 1$.

(xi) If $x > 1 \vee x < 2$ then $x := x + 1$

Sol: For $x = 1$, $x > 1 \vee x < 2$ becomes $1 > 1 \vee 1 < 2$
This shows that, for $x = 1$, $x > 1 \vee x < 2$ is true.
The statement executes, and the value of x changes from x to $x + 1$, i.e. $x = 2$.

(xii) If $x > 1 \wedge x < 2$ then $x := x + 1$

Sol: For $x = 1$, $x > 1 \wedge x < 2$ becomes $1 > 1 \wedge 1 < 2$
This shows that, for $x = 1$, $x > 1 \wedge x < 2$ is false.
The statement does not execute, and the value of x does not change, so $x = 1$.

Q.8 Write the converse, inverse and contrapositive of the following implications:

(i) If $3 > 1$, then $2 < 1$

Sol: If $3 > 1$, then $2 < 1$
Let
$p : 3 > 1$ $\qquad$ $q : 2 < 1$
Then the implication becomes.

$$p \to q : \text{If } 3 > 1, \text{ then } 2 < 1$$

Converse:
Converse of $p \to q$ is

$$q \to p : \text{If } 2 < 1, \text{ then } 3 > 1$$

Inverse:
Inverse of $p \to q$ is

$\neg p \to \neg q$: If $\neg(3 > 1)$, then $\neg(2 < 1)$

Since $\neg(3 > 1) = 3 \le 1$, $\neg(2 < 1) = 2 \ge 1$, therefore, the inverse is

$\neg p \to \neg q$: If $3 \le 1$, then $2 \ge 1$

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \to \neg p$: If $2 \ge 1$, then $3 \le 1$

(ii) If 1 + 1 = 3, then cats can fly.

Sol: If 1 + 1 = 3, then cats can fly.

Let

$p : 1 + 1 = 3$ $\quad$ q : Cats can fly

Then the implication becomes.

$p \to q$: If 1 + 1 = 3, then cats can fly.

Converse:

Converse of $p \to q$ is

$q \to p$: If cats can fly, then 1 + 1 = 3.

Inverse:

Inverse of $p \to q$ is

$\neg p \to \neg q$: If $1 + 1 \neq 3$, then cats cannot fly.

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \to \neg p$: If cats cannot fly, then $1 + 1 \neq 3$

(iii) If I buy a prize bond, I will win a prize.

Sol: If I buy a prize bond, I will win a prize.

Let

p : I buy a prize bond $\quad$ q : I will win a prize

Then the implication becomes.

$p \to q$: If I buy a prize bond, I will win a prize.

Converse:

Converse of $p \to q$ is

$q \to p$: If I win a prize, then I will buy a prize bond.

Inverse:

Inverse of $p \to q$ is

$\neg p \to \neg q$: If I do not buy a prize bond, I will not win a prize.

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \to \neg p$: If I will not win a prize, then I do not buy a prize bond

(iv) It is snowing if it is below freezing.

Sol: It is snowing if it is below freezing.

Let

p : It is below freezing $\quad$ q : It is snowing

Then the implication becomes.

$p \to q$: It is snowing if it is below freezing.

Converse:

Converse of $p \to q$ is

$q \to p$: It is below freezing if it is snowing.

Inverse:

Inverse of $p \to q$ is

$\neg p \to \neg q$: It is not snowing if it is not below freezing.

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \to \neg p$: It is not below freezing if it is not snowing.

(v) **You will qualify interview if you answer all questions.**

Sol: You will qualify interview if you answer all questions.

Let

p : You answer all questions

q : You will qualify interview

Then the implication becomes.

$p \to q$: You will qualify interview if you answer all questions.

Converse:

Converse of $p \to q$ is

$q \to p$: You will answer all questions if you qualify interview.

Inverse:

Inverse of $p \to q$ is

$\neg p \to \neg q$: You will not qualify interview if you do not answer all questions.

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \to \neg p$: You will not answer all questions if you do not qualify interview.

(vi) **You will miss the final examination if you have flue.**

Sol: You will miss the final examination if you have flue.

Let

p : You have flue

q : You will miss the final examination

Then the implication becomes.

$p \to q$: You will miss the final examination if you have flue.

Converse:

Converse of $p \to q$ is

$q \to p$: You will have flue if you miss the final examination.

Inverse:

Inverse of $p \to q$ is

$\neg p \to \neg q$: You will not miss the final examination if you do not have flue.

Contrapositive:

Contrapositive of $p \to q$ is

$\neg q \rightarrow \neg p$: You will not have flue if you do not miss the final examination.

(vii) The home team wins whenever it is raining.

Sol: The home team wins whenever it is raining.

Let

p : It is raining

q : The home team wins

Then the implication becomes.

$p \rightarrow q$: The home team wins whenever it is raining.

Converse:

Converse of $p \rightarrow q$ is

$q \rightarrow p$: It is raining whenever the home team wins.

Inverse:

Inverse of $p \rightarrow q$ is

$\neg p \rightarrow \neg q$: The home team does not win whenever it is not raining.

Contrapositive:

Contrapositive of $p \rightarrow q$ is

$\neg q \rightarrow \neg p$: It is not raining whenever the home team does not win.

(viii) I am going to Murree if I pass this class.

Sol: I am going to Murree if I pass this class.

Let

p : I pass this class

q : I am going to Murree

Then the implication becomes.

$p \rightarrow q$: I am going to Murree if I pass this class.

Converse:

Converse of $p \rightarrow q$ is

$q \rightarrow p$: I pass this class if I am going to Murree.

Inverse:

Inverse of $p \rightarrow q$ is

$\neg p \rightarrow \neg q$: I am not going to Murree if I do not pass this class.

Contrapositive:

Contrapositive of $p \rightarrow q$ is

$\neg q \rightarrow \neg p$: I do not pass this class if I am not going to Murree.

(ix) If it rains today, then I will stay at home.

Sol: If it rains today, then I will stay at home.

Let

p : It rains today

q : I will stay at home

Then the implication becomes.

$p \rightarrow q$: If it rains today, then I will stay at home.

Converse: